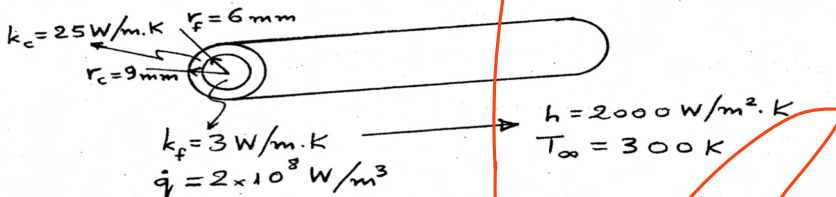


Problem 1: (50 points)

A nuclear reactor fuel element consists of a solid cylindrical pin of radius r_f and thermal conductivity k_f . The fuel pin is in good contact with a cladding material of outer radius r_c and thermal conductivity k_c . Consider steady-state conditions for which uniform heat generation occurs within the fuel at a volumetric rate $\dot{q}$ and the outer surface of the cladding is exposed to a coolant that is characterized by a temperature T_∞ and a convection coefficient h .

- Obtain equations for the temperature distributions $T_f(r)$ and $T_c(r)$ in the fuel and cladding, respectively. Express your results exclusively in terms of the foregoing variables.
- Consider a uranium oxide fuel pin for which $k_f = 3 \text{ W/m}\cdot\text{K}$ and $r_f = 6 \text{ mm}$ and cladding for which $k_c = 25 \text{ W/m}\cdot\text{K}$ and $r_c = 9 \text{ mm}$. If $\dot{q} = 2 \times 10^8 \text{ W/m}^3$, $h = 2000 \text{ W/m}^2\cdot\text{K}$, and $T_\infty = 300 \text{ K}$, what is the maximum temperature in the fuel element?



(a)

$$\frac{1}{r} \frac{d}{dr} \left(k_f r_f \frac{dT_f}{dr_f} \right) + \dot{q} = 0$$

$$\rightarrow \int \frac{d}{dr} \left(r_f \frac{dT_f}{dr} \right) = \int -\frac{\dot{q} r_f}{k_f}$$

$$\rightarrow r_f \frac{dT_f}{dr} = -\frac{\dot{q} r_f^2}{2k_f} + C_1$$

$$\rightarrow \int \frac{dT_f}{dr} = \int -\frac{\dot{q} r_f}{2k_f} + \int \frac{C_1}{r_f} \quad (1)$$

$$\rightarrow T_f(r) = -\frac{\dot{q} r_f^2}{4k_f} + C_1 \ln r_f + C_2 \quad (2)$$

heat equation

$$\frac{1}{r} \frac{\partial}{\partial r} \left(k_c r_c \frac{\partial T_c}{\partial r} \right) = 0 \rightarrow \text{steady state}$$

$$\rightarrow \int d \left(k_c r_c \frac{dT_c}{dr} \right) = \int 0$$

$$\rightarrow k_c r_c \frac{dT_c}{dr} = C_3$$

$$\rightarrow \frac{dT_c}{dr} = \frac{C_3}{k_c r_c} \quad (3)$$

$$\rightarrow T_c(r) = \frac{C_3}{k_c} \ln r_c + C_4 \quad (4)$$

BC 1 : at $r_f = 0$ $\frac{dT_f}{dr} = 0$

$$\rightarrow \underline{C_1} = 0$$

$$\Rightarrow T_f(r) = \frac{-\dot{q} r_f^2}{4 k_f} + C_2$$

BC 2 : $-k_c \frac{dT_c(r)}{dr_c} \Big|_{r=r_c} = h(T_c - T_\infty)$

$$-k_c \left(\frac{C_3}{k_c r_c} \right) = h \left(\frac{C_3}{k_c} \ln r_c + C_4 - T_\infty \right)$$

$$\rightarrow -\frac{C_3}{r_c} = \frac{h C_3}{k_c} \ln r_c + h C_4 - h T_\infty$$

BC 3 : at $r_c = r_f$ $T_c(r) = T_f(r)$

$$\frac{C_3}{k_c} \ln r_f + C_4 = \frac{-\dot{q} r_f^2}{4 k_f} + C_2$$

BC 4 : at $r_c = r_f$ $k_c \frac{dT_c(r)}{dr} = k_f \frac{dT_f(r)}{dr}$

$$k_c \left(\frac{C_3}{k_c r_c} \right) = k_f \left(\frac{-\dot{q} r_f}{2 k_f} \right)$$

$$\rightarrow \underline{C_3} = \frac{-\dot{q} r_f^2}{2}$$

$$\rightarrow \frac{\dot{q} r_f^2}{2 r_c} = \frac{h C_3}{k_c} \ln r_c + h C_4 - h T_\infty$$

$$\rightarrow \frac{\dot{q} r_f^2}{2 r_c} = \frac{h}{k_c} \left(\frac{-\dot{q} r_f^2}{2} \right) \ln r_c + h C_4 - h T_\infty$$

$$\Rightarrow \underline{C_4} = \frac{\dot{q} r_f^2}{2r_c h} + \frac{\dot{q} r_f^2}{2k_c} \ln r_c + T_\infty \quad \text{plug in for BC.3}$$

$$\frac{C_3}{k_c} \ln r_f + C_4 = - \dot{q} \frac{r_f^2}{4k_f} + C_2$$

$$\rightarrow - \frac{\dot{q} r_f^2}{2k_c} \ln r_f + \frac{\dot{q} r_f^2}{2r_c h} + \frac{\dot{q} r_f^2}{2k_c} \ln r_c + T_\infty = - \frac{\dot{q} r_f^2}{4k_f} + C_2$$

$$\Rightarrow \underline{C_2} = \frac{\dot{q} r_f^2}{2k_c} \ln \left(\frac{r_c}{r_f} \right) + \frac{\dot{q} r_f^2}{2r_c h} + T_\infty + \frac{\dot{q} r_f^2}{4k_f}$$

Plug in (2)

at $r = r_f$

$T_f = T_c$

$$T_f(r) = - \frac{\dot{q} r_c^2}{4k_f} + C_1 \ln r_f + C_2$$

$$\rightarrow T_f(r) = - \frac{\dot{q} r_c^2}{4k_f} + \frac{\dot{q} r_f^2}{2k_c} \ln \left(\frac{r_c}{r_f} \right) + \frac{\dot{q} r_f^2}{2r_c h} + T_\infty + \frac{\dot{q} r_f^2}{4k_f}$$

$$\Rightarrow T_f(r) = \frac{\dot{q} r_f^2}{2k_c} \ln \left(\frac{r_c}{r_f} \right) + \frac{\dot{q}}{4k_f} (r_f^2 - r^2) + \frac{\dot{q} r_f^2}{2r_c h} + T_\infty$$

$$\text{in (4)} \quad T_c(r) = \frac{C_3}{k_c} \ln r_f + C_4$$

$$= - \frac{\dot{q} r_f^2}{2k_c} \ln r_f + \frac{\dot{q} r_f^2}{2r_c h} + \frac{\dot{q} r_f^2}{2k_c} \ln r_c + T_\infty$$

$$\Rightarrow T_c(r) = \frac{\dot{q} r_f^2}{2r_c h} + \frac{\dot{q} r_f^2}{2k_c} \ln \left(\frac{r_2}{r_1} \right) + T_\infty$$

(b)

at $r = 0$

$T = T_{\max}$

$$T_f(r)_{\max} = \frac{\dot{q} r_f^2}{2k_c} \ln \left(\frac{r_c}{r_f} \right) + \frac{\dot{q}}{4k_f} (r_f^2 - r^2) + \frac{\dot{q} r_f^2}{2r_c h} + T_\infty$$

$$= \frac{(2 \times 10^8)(0.006)^2}{(2)(25)} \ln \left(\frac{0.009}{0.006} \right) + \frac{(2 \times 10^8)(0.006^2)}{(4)(3)} + \frac{(2 \times 10^8)(0.006)^2}{(2)(0.009)(2000)} + 300$$

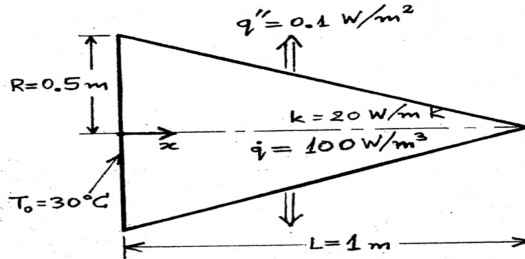
$$T_f(r)_{\max} = 58.38 + 600 + 200 + 300$$

$$T_f(r)_{\max} = 1158.38 \text{ K}$$

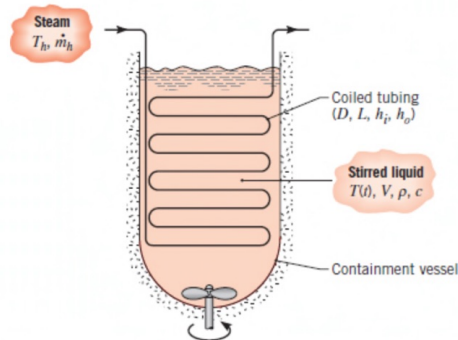
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Problem 2: (50 points)

A solid cone as shown is of circular cross section and the radius of its base is 0.5 m. The cone of length 1 m and thermal conductivity $20 \text{ W/m}\cdot\text{K}$ has uniform volumetric heat generation of 100 W/m^3 . A constant heat flux of 0.1 W/m^2 is leaving the cone through the lateral surface of the cone, while the end surface at $x = 0$ is maintained at $T_o = 30^\circ\text{C}$. Determine the temperature profile $T(x)$ and the temperature $T(L)$.



5.16 Batch processes are often used in chemical and pharmaceutical operations to achieve a desired chemical composition for the final product and typically involve a **transient heating operation** to take the product from room temperature to the desired process temperature. Consider a situation for which a chemical of density $\rho = 1200 \text{ kg/m}^3$ and specific heat $c = 2200 \text{ J/kg} \cdot \text{K}$ occupies a volume of $V = 2.25 \text{ m}^3$ in an insulated vessel. The chemical is to be heated from room temperature, $T_i = 300 \text{ K}$, to a process temperature of $T = 450 \text{ K}$ by passing saturated steam at $T_h = 500 \text{ K}$ through a coiled, thin-walled, 20-mm-diameter tube in the vessel. Steam condensation within the tube maintains an interior convection coefficient of $h_i = 10,000 \text{ W/m}^2 \cdot \text{K}$, while the highly agitated liquid in the stirred vessel maintains an outside convection coefficient of $h_o = 2000 \text{ W/m}^2 \cdot \text{K}$.



If the chemical is to be heated from 300 to 450 K in 60 min, what is the required length L of the submerged tubing?

$$-\dot{E}_{\text{out}} = \dot{E}_{\text{st}}$$

$$\rightarrow -h A_s (T - T_{\infty}) = \rho V c \frac{dT}{dt}$$

$$[\theta = T_h - T_{\infty}]$$

$$\rightarrow -h A_s \theta = \rho V c \frac{d\theta}{dt}$$

$$\rightarrow -\theta = \frac{\rho V c}{h A_s} \frac{d\theta}{dt}$$

$$\frac{d\theta}{dt} = \frac{dT}{dt} \quad (T_{\infty} \text{ is constant})$$

$$\rightarrow -\int_0^t dt = \frac{\rho V c}{h A_s} \int_{\theta_i}^{\theta} \frac{d\theta}{\theta}$$

$$\rightarrow -\left[t \right]_0^t = \frac{\rho V c}{h A_s} \left[\ln \theta \right]_{\theta_i}^{\theta}$$

$$\rightarrow -t = \frac{\rho V_c}{h A_s} (\ln \theta - \ln \theta_i)$$

$$\rightarrow t = \frac{\rho V_c}{h A_s} \cdot \ln\left(\frac{\theta_i}{\theta}\right)$$

$$\Rightarrow \frac{1}{A_s} = \frac{t h}{\rho V_c \ln\left(\frac{\theta_i}{\theta}\right)}$$

$$\left[\begin{aligned} h &= \frac{1}{\left(\frac{1}{h_o}\right) + \left(\frac{1}{h_i}\right)} = \frac{1}{\left(\frac{1}{2000}\right) + \left(\frac{1}{10,000}\right)} \\ \Rightarrow h &= 1666.67 \text{ W/m}^2 \cdot \text{K} \end{aligned} \right]$$

$$\Rightarrow \frac{1}{A_s} = \frac{(3600 \cancel{s}) (1666.67 \cancel{\text{W}}/\text{m}^2 \cdot \cancel{\text{K}})}{1200 \left(\frac{\text{kg}}{\text{m}^3}\right) \left(2.25 \cancel{\text{m}^3}\right) \left(2200 \frac{\cancel{\text{J}}}{\text{kg}}\right) \ln\left(\frac{500 - 300}{500 - 450}\right)}$$

$$\Rightarrow \frac{1}{A_s} = 0.7286$$

$$\Rightarrow A_s = 1.3724 \text{ m}^2$$

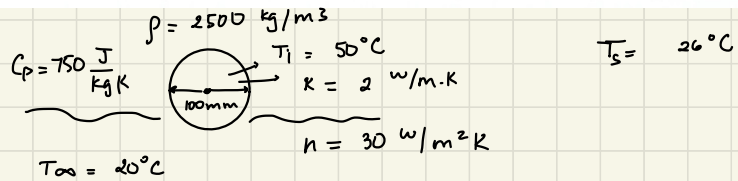
$$A_s = \pi D L$$

$$\Rightarrow L = \frac{A_s}{\pi D} = \frac{1.3724 \text{ m}^2}{\pi (0.02 \text{ m})}$$

$$\Rightarrow \boxed{L = 21.84 \text{ m}}$$



5.58 A $D = 100 \text{ mm}$ diameter solid sphere, initially at a uniform temperature of $T_i = 50^\circ\text{C}$, is placed in a flowing fluid at $T_\infty = 20^\circ\text{C}$. The sphere properties are $k = 2 \text{ W/m} \cdot \text{K}$, $\rho = 2500 \text{ kg/m}^3$, and $c_p = 750 \text{ J/kg} \cdot \text{K}$. The convection heat transfer coefficient is $h = 30 \text{ W/m}^2 \cdot \text{K}$. At what time does the sphere's surface temperature reach 26°C ? How much energy has been transferred from the sphere at this time?



$$Bi = \frac{h_a r_o}{3k} = \frac{(30)(0.050)}{3(2)} = 0.25$$

lumped capacitance is not appropriate

$$Bi = \frac{h_a r_o}{k} = \frac{(30)(0.050)}{2} = 0.75$$

$$F_o = -\frac{1}{\delta_1^2} \ln \left[\frac{\theta_o^*}{C_1} \right] = -\frac{1}{\delta_1^2} \ln \left[\frac{1}{C_1} + \frac{T_s - T_\infty}{T_i - T_\infty} \right]$$

$$\delta_1 = \frac{1.4320 + 1.3525}{2} = 1.39225$$

$$C_1 = \frac{1.2236 + 1.1978}{2} = 1.2107$$

$$\theta^* = C_1 e^{(-\delta_1^2 F_o)} \frac{1}{\delta_1 r^*} \sin(\delta_1 r^*)$$

$$\frac{T - T_\infty}{T_i - T_\infty} = \frac{1.2107}{1.39225} e^{(-1.39225^2 F_o)} \sin(1.39225)$$

$$\frac{1}{5} = 0.8558 e^{-1.39225^2 F_o}$$

$$0.2337 = e^{-1.39225^2 F_o}$$

$$\rightarrow F_o = 0.7410$$

$$F_0 = \frac{t \alpha}{r_0^2} \Rightarrow t = \frac{F_0 r_0^2}{\alpha}$$

$$\left[\alpha = \frac{k}{\rho c p} = \frac{2}{(2500)(150)} = 1.067 \times 10^{-6} \right]$$

$$\Rightarrow t = \frac{(0.7410)(0.05^2)}{1.067 \times 10^{-6}} = \boxed{1736.2 \text{ s}}$$

$$\text{Eq. 5.55} \quad \frac{Q}{Q_0} = 1 - \frac{3\theta_0^*}{\delta_1^3} [\sin(\delta_1) - \delta_1 \cos \delta_1]$$

$$[Q_0 = \rho c V (T_i - T_\infty)]$$

$$Q = \left[1 - \frac{3c_1}{\delta_1^3} e^{(-\delta_1^2 F_0)} \frac{1}{\delta_1} \sin(\delta) \right] [\sin(\delta_1) - \delta_1 \cos(\delta_1)]$$

$$[\rho c V (T_i - T_\infty)]$$

$$Q = \left[1 - \frac{3(1.2107)}{(1.39225)^4} e^{(-1.39225)^2 (0.7410)} \sin(1.39225) \right]$$

$$\left[\sin(1.39225) - 1.39225 \cos(1.39225) \right]$$

$$\times (2500)(150) \left[\frac{4}{3} \pi \left[\frac{(0.1)}{2} \right]^3 \right] (50 - 20)$$

$$Q = (0.7738)(0.7368)(9375 \pi)$$

$$\boxed{Q = 16,791.89 \text{ J}}$$